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EXTERNAL GPU

UNFINISHED LECTURE NOTES

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These notes are, by their very nature, unfinished, and they improve with every reader. If you spot an error, disagree with a framing, or want to add a source, a question, or a position, open a pull request at https://github.com/Quillstacks/LectureMaterial/tree/main/lecturenotes/notes_egpu. Every contribution is welcome.



Quickstart

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Aaaah-ah-ah-ah, aaaah-ah-ah-ah, Thunder!

LUCKY YOU. Your machine is compatible. Ask for the box next. You get a time slot and hand the box back with its cable and power cord. Never attempt to open the case, and don't drop it.

PREPARE THE LAPTOP BEFORE PICKING UP THE BOX. I want to see it running before you will be allowed to take it with you. On Windows, install pending updates, the newest BIOS from your laptop's manufacturer, and the NVIDIA driver from NVIDIA's site; outdated firmware can hide the card ¹. On Ubuntu, run `sudo ubuntu-drivers install`.

PLUG IN. Use the cable that comes with the box, connect the power cord, and flip the switch on the back. The front LED lights once the laptop is on. Keep the ventilator shafts clear.

APPROVE THE DEVICE. Windows on Thunderbolt 3 asks once, choose Always connect. On Ubuntu, run `boltctl enroll` with the device id that `boltctl list` shows.

CHECK THE CARD. `nvidia-smi` in a terminal shows it with its 16 GB.

ALWAYS EJECT BEFORE YOU UNPLUG. Stop every process `nvidia-smi` lists, then on Windows use the NVIDIA eject icon in the taskbar tray. Pulling the cable from a running card can crash the laptop.

THE SOUNDTRACK for first use. Meant to be played loud. Your neighbours will understand.



SCAN THE CODE and send a mail stating your name, course, laptop model, why and what you want to run.



¹ Sonnet Technologies. egpu breakaway box graphics guide for windows 10 and 11. https://www.sonnettech.com/support/downloads/manuals/windows_egpu_info.pdf, 2023

CHECKS. Nothings in Device Manager means the driver is missing. Nothing at all means the box is not detected: Check cable, power switch, and port.

AFTER A RECONNECT on Windows, click Connect GPU in the same tray icon.